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Original Article

Prevalence, molecular characterization, and antimicrobial resistance profile of *Clostridium perfringens* from India: A scoping review

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ABSTRACT

Clostridium perfringens is one of the most important foodborne pathogens that causes histotoxic diseases and intestinal infections in both humans and animals. The present scoping review has been designed to analyze the literature published during 2000–2021 from India on the prevalence, molecular characterization, and antimicrobial resistance profile of *C. perfringens* isolates recovered from humans, animals, animal-based foods, and associated environmental samples. The peer-reviewed articles retrieved from four electronic databases (Google Scholar, PubMed, Science Direct, and Web of Science) were assessed using PRISMA-ScR guidelines. A total of 32 studies from India were selected on the basis of their relevance and inclusion criteria. The overall prevalence of *C. perfringens* among domestic animals having history of clinical symptoms and among healthy animals was found to be 65.8% (508/772) and 42.8% (493/1152), respectively. The pathogen was also detected in clinically affected wild animals (75%), healthy wild animals (35.4%), and captive birds (24.5%). The detection of *C. perfringens* among poultry having necrotic enteritis and among healthy birds was found to be 66.8% (321/480) and 25.6% (80/312), respectively. The detection of pathogen among animal-based foods (i.e., meat, milk, and fish and their products) and environmental samples depicted a prevalence of 20.8% (325/1562) and 30.2% (23/76), respectively. However, the prevalence of *C. perfringens* among humans having history of diarrhea and among healthy humans was found to be 25% (70/280) and 23.2% (36/155), respectively. The genotyping of *C. perfringens* isolates revealed that toxin type A was found to be the most prevalent genotype. Along with the alpha toxin gene (*cpa*), beta (*cpb*), epsilon (*etx*), iota (*itx*), enterotoxin (*cpe*), beta-2 toxin (*cpb2*), and NetB (*netB*) toxins were also detected in different combinations. Antimicrobial resistance profile of *C. perfringens* isolates recovered from different sources demonstrated that the highest resistance was detected against sulphonamides (76.8%) and tetracycline (41.3%) by phenotypic and genotypic detection methods, respectively. Comprehensive scientific studies covering different geographical areas at the human-animal-environment interface are crucial to generalize the real magnitude of *C. perfringens*-associated problem in India and for establishing a reliable database.

1. Introduction

Clostridium perfringens is an anaerobic, gram-positive, and spore-forming bacterium, having ubiquitous environmental presence in soil, decaying vegetation, feces, and the intestinal tracts of humans, animals, and birds [1,2]. The organism is responsible for many histotoxic and enterotoxic diseases in humans and animals [3,4]. It has a rapid doubling potential with generation time of less than 10 min under optimal growth conditions [5], which allow this bacterium to quickly reach pathogenic burden in foods, wounds, or in the intestine [2,6]. The gliding motility of this bacterium contributes to the potentially virulence-related functions like adherence and biofilm formation [7]. The virulence of this bacterium can be attributed by its armory of more than 20 potent toxins and hydrolytic enzymes [2,8]. Based on the toxin production patterns of

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different strains, the *C. perfringens* isolates can be divided into seven toxin types (A-G) [6,9]. These toxin types are generally involved in specific disease syndromes [2,9,10]. The alpha toxin (phosholipase-C) encoded by the *cpa* gene is produced by all the toxinotypes of *C. perfringens* [6,9]. Among different toxinotypes, types A, C, and F are commonly involved in human infections, whereas types B, D, E, and G are of major importance in veterinary medicine [2,11]. Besides these toxin-genotypes, it may also carry other extracellular toxic molecules such as enterotoxin (CPE) and beta2 toxin (CPB2) [12]. CPE is chromosome or pCW3-like plasmid-mediated toxin and is associated with large bowel diarrhea and gastrointestinal syndromes in humans and animals [2,13]. About 1–5% of *C. perfringens* type A strains isolated from humans and animals were found to carry the *C. perfringens* enterotoxin (*cpe*) gene [14,15]. *C. perfringens* enterotoxin (CPE) producing type A is the second most common cause of foodborne illness in the United States (US), where they cause nearly 1 million cases per annum, resulting in a net economic loss of US $382 million [16–18]. Enterotoxigenic *C. perfringens* type A is associated with antibiotic-associated diarrhea (AAD) (approximately 5–15% of all cases of AAD) and sporadic diarrhea (SD) [19–21]. Toxin-genotypes C, D, and E are also known to produce the CPE [2,9]. CPE-producing *C. perfringens* toxin-genotypes are typically transmitted to humans via contaminated and improperly cooked foods [22]. Except for type F strains which carry the chromosomal *cpe* gene, *C. perfringens* strains usually need to acquire a plasmid-borne toxin gene to cause gastrointestinal diseases [2]. Toxin CPB2 is pCW3- or pCP13-like plasmid-mediated necrotizing and lethal toxin, produced by all genotypes of *C. perfringens*, and is associated with gastrointestinal diseases in humans, pigs, horses, and other animals [8,12,23–25].

The organism causes food poisoning incidences with the symptoms similar with many other foodborne illnesses. Therefore, diagnostic confirmation requires culture of the pathogen. The cultural method is considered to be gold standard for the detection of *C. perfringens*, but they have their own limitations as they are time-consuming (require approx. 4–6 days) and labor-intensive [26]. Nucleic acid-based molecular assays such as PCR, real-time PCR and loop-mediated isothermal amplification (LAMP) enables rapid detection of the pathogen and serve as an alternative to culture-based methods [27]. PCR-based technology is considered to be a convenient and highly reliable tool for molecular detection of all the major toxin genes such as alpha (*cpa*), beta (*cpb*), epsilon (*etx*), and iota (*iap*) [28].

Worldwide, indiscriminate and long-term use of antimicrobial agents for growth promotion, prophylactic, and therapeutic purposes have led to the rapid development of antimicrobial resistance among *C. perfringens* isolates [29,30]. In addition, enteric fecal flora from food-producing animals such as poultry and swine can transfer antimicrobial resistance genes to human pathogens via the food chain [31,32]. The emerging problem of transfer of antimicrobial resistance elements between pathogenic and commensal bacteria is also a serious concern [33,34]. Antimicrobial resistance conferred by various genes plays a major role in the transfer of resistance to other bacteria within or between genera [11]. There are reports on resistant antibiogram of *C. perfringens* isolates recovered from different sources in India [11,18,30,35–40].

With this background, the present scoping review analyzes the literature published during 2000–2021 from India on the prevalence, molecular characterization and antimicrobial resistance profile of *C. perfringens* isolates recovered from humans, animals, animal-based foods, and associated environmental samples.

2. Material and methods

2.1. Scoping review approach and eligibility criteria

The present review was conducted by adopting the framework described by Arksey and O'Malley for conducting scoping reviews [41] with minor modifications, including those suggested by Levac et al. [42]. The reporting of the present study is in concordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement extension for scoping reviews (PRISMA-ScR) [43]. The main focus of the present review was on peer-reviewed literature; thus, no grey literature sources were searched.

2.2. Search strategy and data collection

In the present study, literatures published between the years 2000 and 2021 were retrieved from four electronic databases, viz., Google Scholar, PubMed, Science Direct, and Web of Science. The search was conducted during the period from May 15, 2021 to August 20, 2021. The search algorithms consisted of various combinations of keywords, *viz.*, *Clostridium perfringens*, India, meat, milk, fish, sea foods, diarrhea, necrotic enteritis, gastro-enteritis, food poisoning, antibiotic-associated diarrhea, sporadic diarrhea, healthy, humans, animals, and environment.

2.3. Data screening and extraction

The preliminary literature search was carried out for titles, abstracts/summary, and keywords in the first round by three independent researchers. The studies were selected for full text evaluation, if they met the inclusion criteria, i.e., any research article published in between 2000 and 2021 from India on *C. perfringens*, which includes prevalence, molecular characterization (targeting virulence genes, *i.e*., *cpa*, *cpb*, *etx*, *itx*, *cpe*, *cpb2* and *netB*) and antimicrobial resistance profile (phenotypic and/or genotypic characterization).

The selected studies were checked for relevance and further in-depth screening of the full text in the second round. The full-text characterization includes key characteristics such as, publication details, source(s) of *C. perfringens* isolation, prevalence, targeted virulence genes, molecular methods used for characterization of isolates, and antimicrobial resistance profile of *C. perfringens* isolates using phenotypic and genotypic methods.

2.4. Review management

All the steps were reviewed by three independent reviewers. The relevant data screening, collection, confirmation, and data extraction was done through spreadsheets (Excel, Microsoft Office 365, Microsoft Corporation, USA) (Fig. 1). All discrepancies were discussed and resolved through consensus between the reviewers.

2.5. Data analysis

The data on the source(s) of *C. perfringens* isolates, associated virulence genes, and methodology used for the determination of antimicrobial resistance profile of *C. perfringens* isolates were assessed for relevant descriptive statistics using Microsoft Excel (Excel, Microsoft Office 365, Microsoft Corporation, USA). *C. perfringens* isolates carrying the *cpe*/*cpe* + *cpb2* genes in addition to the *cpa* gene (*cpa* + *cpe*/*cpa* + *cpe* + *cpb2*) and classified under type A isolates in previous published literatures were considered under type F in this study according to newer classification of *C. perfringens* toxin types A-G [9].

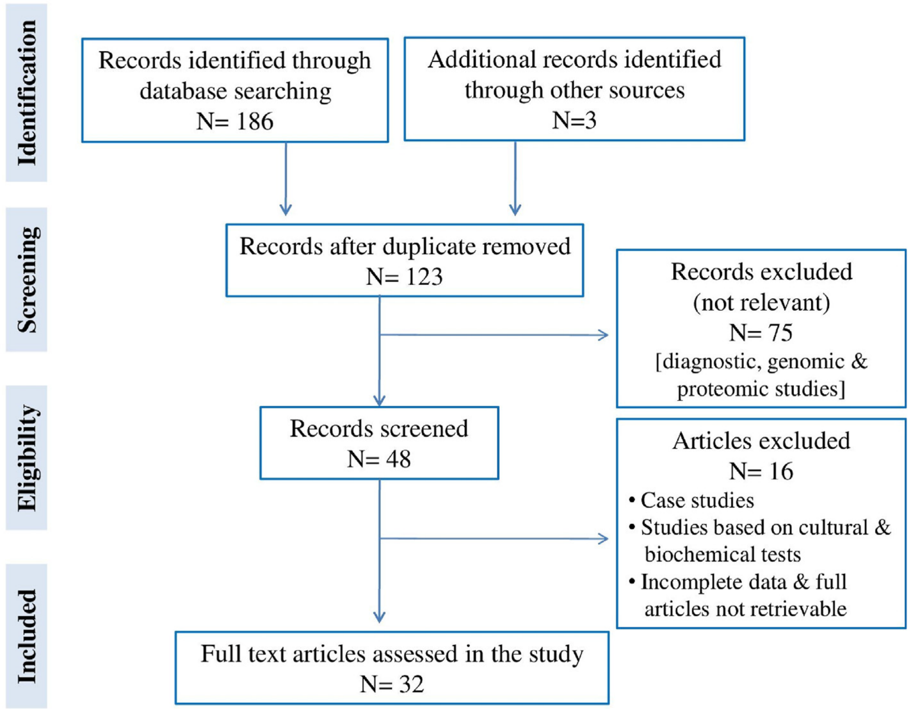

Fig. 1. PRISMA flow chart for the selection of studies included in the scoping review.

3. Results and discussion

A total of 32 studies from India were selected on the basis of their relevance and inclusion criteria, *i.e.*, prevalence, molecular characterization of *C. perfringens* isolates into toxin types A-G by considering different virulence genes (*cpa*, *cpb*, *etx*, *itx*, *cpe*, *cpb2*, and *netB*) and antimicrobial resistance profile (phenotypic and genotypic) of the isolates. The articles on *C. perfringens* from India which were excluded/not considered in this study were either case studies/reports or the studies that did not follow the standard methodology or reporting of unclear results or the studies not based on epidemiology, molecular characterization, and antimicrobial resistance profile of *C. perfringens* isolates. The selected articles (n = 32) were reported from 10 states and 2 union territories (UTs) of India (out of total 28 states and 8 UTs). The regional distribution of selected studies was maximum from Southern states (n = 11), followed by Northern (n = 6), North-Eastern (n = 4), Eastern [n = 2), and Western (n = 1) states of India. Two UTs from which the studies on *C. perfringens* have been reported were Jammu & Kashmir (n = 6) and New Delhi (n = 2). Out of 32 studies, 26 studies were carried out in veterinary colleges/universities; 3 studies were from life science/defense food research laboratories; 1 study was from medical university, and 2 studies were found in collaboration of veterinary and medical professionals.

3.1. Toxinotyping of *C. perfringens* isolates from India

3.1.1. *C. perfringens* isolates recovered from clinically affected and healthy animals and birds

The overall prevalence of *C. perfringens* among animals (bovines, sheep, goat, pig, and dog) and poultry having history of clinical symptoms (diarrhea/enterotoxemia/lamb dysentery/necrotic enteritis) was found to be 65.8% (508/772) and 66.8% (321/480), respectively (Table 1; Supplementary Table 1). On species-wise detection of *C. perfringens* infection in animals, the maximum prevalence was found in sheep (80.1%), followed by that in goat (66.6%), pig (65.4%), bovines (43%), and dogs (25.9%) (Table 1). Toxinotyping of *C. perfringens* isolates (n = 351) of sheep revealed the detection of alpha (*cpa*), beta (*cpb*), epsilon (*etx*), beta2 (*cpb2*), and enterotoxin (*cpe*) genes in various combinations indicating the presence of *C. perfringens* types A (70.9%) as a predominant toxin type, followed by type D (23.9%), type C (4.5%), and type B (0.6%) (Table 1). *C. perfringens* isolates (n = 55) of bovines were predominant with toxin type A (67.3%), followed by type C (10.9%), type F (9.1%), type D (7.3%) and type B (5.4%). Additionally, *C. perfringens* isolates (n = 28) of bovines recovered from different clinical cases were grouped in toxin type A (25%), type C (14.3%), and type F (60.7%). *C. perfringens* isolates (n = 46) of goat were grouped in toxin type A (84.8%), type B (2.2%), type D (4.3%), and type F (8.7%). *C. perfringens* isolates (n = 27) recovered from goats having different disease symptoms were grouped as toxin type A (22.2%), type C (14.8%), and type F (63%). Likewise, *C. perfringens* isolates of pigs (n = 34) were grouped into toxin type A (85.3%), type C (2.9%), and type F (11.8%). Additionally, *C. perfringens* isolates (n = 11) recovered from pigs having different disease symptoms were grouped as type C (63.6%) and type F (36.4%). *C. perfringens* isolates (n = 22) recovered from dogs having history of gastroenteritis were grouped in toxin type A (90.9%), type C (4.5%), and type F (4.5%). However, *C. perfringens* isolates (n = 321) of poultry were grouped in toxin type A (93.8%), type C (4.7%), and type G (1.5%). None of the *C. perfringens* isolates detected from animals and poultry were found positive for type E, *i.e.*, all the isolates were negative for the iota toxin gene. Toxinotyping for the *netB* gene (type G) was carried out in poultry and dog samples, but detected only in poultry samples (Table 1).

The overall prevalence of *C. perfringens* among healthy animals (sheep, bovines, goat, and pig) and poultry was found to be 42.8% (493/1152) and 25.6% (80/312), respectively (Table 2; Supplementary Table 1). On species-wise detection of *C. perfringens* infection, the maximum prevalence was found in pigs (60%),

Table 1
Prevalence and toxinotyping of *C. perfringens* isolates recovered from clinically affected animals and birds.

Species	Prevalence of *C. perfringens* (%)	Toxinotypes, numbers (%)														References
		Type A		Type B	Type C				Type D			Type E	Type F		Type G	
		cpa	*cpa* + *cpb2*	(*cpa* + *cpb* + *etx*)	(*cpa* + *cpb*)	(*cpa* + *cpb* + *cpb2*)	(*cpa* + *cpb* + *cpe*)	(*cpa* + *cpb* + *cpb2*+*cpe*)	(*cpa* + *etx*)	(*cpa* + *etx* + *cpb2*)	(*cpa* + *etx* + *cpe*)	(*cpa* + *itx*)	(*cpa* + *cpe*)	(*cpa* + *cpe* + *cpb2*	(*cpa* + *netB*)	
Sheep	351/438 (80.1)	249/351 (70.9)	0 (0)	2/351 (0.6)	16/351 (4.5)	0 (0)	0 (0)	0 (0)	78/351 (22.2)	4/351 (1.1)	2/351 (0.6)	0 (0)	0 (0)	0 (0)	ND*	[45,46,66,87–90]
Bovine	55/128 (43)	37/55 (67.3)	0 (0)	3/55 (5.4)	5/55 (9.1)	0 (0)	1/55 (1.8)	0 (0)	4/55 (7.3)	0 (0)	0 (0)	0 (0)	4/55 (7.3)	1/55 (1.8)	ND	[22,66]
	NA**	7/28 (25)	0 (0)	NA	0 (0)	3/28 (10.7)	1/28 (3.6)	0 (0)	NA	NA	NA	NA	15/28 (53.6)	2/28 (7.1)	NA	[91]
Goat	46/69 (66.6)	23/46 (50)	16/46 (34.8)	1/46 (2.2)	0 (0)	0 (0)	0 (0)	0 (0)	2/46 (4.3)	0 (0)	0 (0)	0 (0)	2/46 (4.3)	2/46 (4.3)	ND	[18,45,66]
	NA	6/27 (22.2)	0 (0)	NA	1/27 (3.7)	1/27 (3.7)	2/27 (7.4)	0 (0)	NA	NA	NA	NA	15/27 (55.5)	2/27 (7.4)	NA	[91]
Pig	34/52 (65.4)	3/34 (8.8)	26/34 (76.5)	0 (0)	1/34 (2.9)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2/34 (5.9)	2/34 (5.9)	ND	[18]
	NA	0 (0)	0 (0)	NA	1/11 (9.1)	4/11 (36.4)	2/11 (18.2)	0 (0)	NA	NA	NA	NA	4/11 (36.4)	0 (0)	NA	[91]
Dog	22/85 (25.9)	20/22 (91)	0 (0)	0 (0)	1/22 (4.5)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1/22 (4.5)	0 (0)	0 (0)	[11,66]
Poultry	321/480 (66.9)	153/321 (47.7)	148/321 (46.1)	0 (0)	15/321 (4.7)	0 (0)	ND	ND	0 (0)	ND	ND	0 (0)	ND	ND	5/321 (1.5)	[37,39,65,66,92]

ND*- Not done; NA**- Not available.

Table 2
Prevalence and toxinotyping of *C. perfringens* isolates recovered from healthy animals and birds.

Species	Prevalence of *C. perfringens* (%)	Toxinotypes, numbers (%)														References
		Type A		Type B	Type C				Type D			Type E	Type F		Type G	
		cpa	*cpa* + *cpb2*	(*cpa* + *cpb* + *etx*)	(*cpa* + *cpb*)	(*cpa* + *cpb* + *cpb2*)	(*cpa* + *cpb* + *cpe*)	(*cpa* + *cpb* + *cpb2*+*cpe*)	(*cpa* + *etx*)	(*cpa* + *etx* + *cpb2*)	(*cpa* + *etx* + *cpe*)	(*cpa* + *itx*)	(*cpa* + *cpe*)	(*cpa* + *cpe* + *cpb2*	(*cpa* + *netB*)	
Sheep	197/350 (56.3)	139/197 (70.5)	0 (0)	0 (0)	1/197 (0.5)	0 (0)	0 (0)	0 (0)	48/197 (24.4)	8/197 (4.1)	1/197 (0.5)	0 (0)	0 (0)	0 (0)	ND*	[45,46,66,87,90]
Bovine	235/672 (35)	135/235 (57.4)	20/235 (8.5)	5/235 (2.1)	31/235 (13.2)	1/235 (0.4)	5/235 (2.1)	0 (0)	9/235 (3.8)	0 (0)	3/235 (1.3)	0 (0)	19/235 (8.1)	7/235 (3)	ND	[22,66,100]
	NA**	0 (0)	0 (0)	NA	0 (0)	0 (0)	1/6 (16.6)	0 (0)	NA	NA	NA	NA	1/6 (16.6)	4/6 (66.6)	NA	[91]
Goat	31/80 (38.7)	21/31 (67.7)	4/31 (12.9)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	3/31 (9.7)	1/31 (3.2)	1/31 (3.2)	0	1/31 (3.2)	0	ND	[18,45,66]
	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	0 (0)	0 (0)	NA	NA	NA	NA	1/5 (20)	4/5 (80)	NA	[91]
Pig	30/50 (60)	16/30 (53.3)	13/30 (43.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)		1/30 (3.3)	ND	[18]
	NA	0 (0)	0 (0)	NA	0 (0)	0 (0)	0 (0)	0 (0)	NA	NA	NA	NA	1/2 (50)	1/2 (50)	NA	[91]
Poultry	80/312 (25.6)	44/80 (55)	12/80 (15)	0 (0)	3/80 (3.7)	14/80 (17.5)	0 (0)	1/80 (1.2)	0 (0)	0 (0)	0 (0)	0 (0)	1/80 (1.2)	5/80 (6.2)	0 (0)	[40,66,92]

ND*- Not done; NA**- Not available.

followed by that in sheep (56.3%), goat (38.7%), and bovines (35%) (Table 2). *C. perfringens* isolates (n = 30) of pig were grouped in toxin type A (96.6%) and type F (3.3%). Additionally, 2 isolates recovered from feces of healthy pigs were grouped in toxin type F. Likewise, *C. perfringens* isolates (n = 197) of sheep were grouped in toxin type A (70.5%), type C (0.5%), and type D (28.9%). *C. perfringens* isolates (n = 31) of goats were grouped in type A (80.6%), type D (16.1%), and type F (3.2%). Additionally, *C. perfringens* isolates (n = 5) recovered from feces/rectal swabs of healthy goats were grouped in toxin type F. *C. perfringens* isolates (n = 235) of bovines were grouped in toxin type A (65.9%), type B (2.1%), type C (15.7%), type D (5.1%), and type F (11.1%). Additionally, *C. perfringens* isolates (n = 6) recovered from feces/rectal swabs/udder swab of healthy bovines were grouped in type C (16.6%) and type F (83.3%). However, *C. perfringens* isolates (n = 80) of poultry recovered from feces/cloacal swabs/intestinal scrapings were predominant with toxin type A (70%), followed by type C (22.5%) and type F (7.5%). Similarly, as in clinically affected animals and poultry, none of the *C. perfringens* isolate recovered from healthy animals and poultry was found positive for toxin type E. Toxinotyping for the *netB* gene (type G) was carried out in poultry samples, and none were found positive for this gene (Table 2).

Toxinotyping of *C. perfringens* isolates was carried out in animals and birds having either unknown health status [11] or isolates recovered from a mixed population of birds (healthy and diseased) [44]. Anju et al. [11] reported 92.6% and 88.2% prevalence of *C. perfringens* in feces/intestinal contents of domestic animals and birds, respectively. *C. perfringens* isolates (n = 25) recovered from domestic animals were grouped into toxin type A (80%; *cpa*), type C (4%; *cpa* + *cpb*), type D (12%; *cpa* + *etx*), and type F (4%; *cpa* + *cpe*). However, birds' isolates (n = 30) were grouped only in toxin type A (*cpa* and *cpa* + *cpb2*). Dar et al. [44] reported 28.1% prevalence of *C. perfringens* in feces of poultry, and all the isolates were grouped under toxin type A (*cpa* and *cpa* + *cpb2*).

C. perfringens toxinotypes are responsible for varied disease syndromes in livestock and poultry [45]. Though *C. perfringens* was considered as part of normal intestinal flora, whenever there is an unfavorable alteration in intestinal environment either by sudden changes in diet or other factors, the organism proliferates in large numbers and produces several potent toxins, resulting in various disease symptoms [1,46]. Most of the *C. perfringens* isolates recovered from the selected studies were type A, which was in accordance with earlier reports on the global dominance of type A in various animal species and poultry [47–51]. Beta toxin is considered as the main virulence factor in type C infections [25], which is very sensitive to trypsin digestion. Animals with low levels of intestinal trypsin (lambs) are usually the most susceptible groups for beta toxin-producing *C. perfringens* isolates [25,52]. Type D isolates of *C. perfringens* may cause enterotoxemia in sheep, goats, and cattle [2,9,53]. However, type D isolates of *C. perfringens* were also detected in healthy animals in the present study. Similarly, the detection of toxin type D in healthy animals was also reported earlier [53–55]. The non-detection of toxin type E in any of the samples of animals and poultry have been found to be concurrent with earlier studies [50,51,56,57]. *C. perfringens cpe* and *cpb2* toxin genes have been widely reported as potential contributors to enteric diseases [45]. Along with the *cpa* gene, *C. perfringens* isolates recovered from various animal species also carried *cpb2* and *cpe* genes, indicating involvement of these toxin genes in gastrointestinal diseases of animals. The presence of the *cpb2* and *cpe* genes have been reported from various animal species in earlier studies [37,54,58,59]. However, there is no molecular relationship of the *cpb2* and *cpe* genes with enteritis in animals has been established, but evidence shows that the presence of *cpb2* and *cpe* gene is usually higher in animals with gastrointestinal symptoms as compared to healthy animals [11].

The higher prevalence of *C. perfringens cpb2* gene in the suspected cases of necrotic enteritis as compared to that in healthy birds might indicate its probable role in the development of necrotic enteritis. These results were concurrent with earlier study reports [44,60,61]. Detection of the *netB* gene along with the *cpa* gene of *C. perfringens* in the necrotic enteritis-suspected birds point towards possible involvement of these genes in the causation of disease [49,62–64]. The absence of additional virulence-associated gene(s) in the *C. perfringens* type C isolates indicates their rare association with cases of necrotic enteritis in birds [39,65].

3.1.2. *C. perfringens isolates recovered from captive wild animals and birds*

Limited studies have been carried out for the detection of *C. perfringens* in captive wild animals and birds from India [66,67]. The overall prevalence of *C. perfringens* in clinically affected wild animals, healthy wild animals and birds was found to be 75%, 35.4%, and 24.5%, respectively (Table 3; Supplementary Table 1). Rahman et al. [66] reported 75%, 37.5%, and 75% prevalence of *C. perfringens* in clinically affected wild animals, healthy wild animals, and birds, respectively (Supplementary Table 1). Rahman et al. [66] screened all the samples only for the presence of the *cpa* gene of *C. perfringens*; hence, all the isolates were categorized under toxin type A. Milton et al. [67] reported 34.9% and 22.5% prevalence of *C. perfringens* in healthy wild animals and birds, respectively (Supplementary Table 1). *C. perfringens* isolates of wild animals (n = 74) were grouped into toxin type A (97.3%) and type B (2.7%). However, all the isolates (n = 23) of captive wild birds were grouped under toxin type A (Supplementary Table 1) [67]. Similarly, earlier studies from different parts of the globe also reported the predominance of toxin type A in *C. perfringens* isolates recovered from wild animals and birds [68–71].

3.1.3. *C. perfringens isolates recovered from foods of animal origin*

The overall prevalence of *C. perfringens* among foods of animal origin (meat, milk, fish and their products) was found to be 20.8% (325/1562) (Table 4; Supplementary Table 2). Almost similar prevalence of *C. perfringens* was reported in different animal-based foods (meat and meat products- 19.2%; milk and milk products- 20%; fish and fish products- 21.5%). *C. perfringens* isolates (n = 82) recovered from meat and meat products were grouped into toxin type A (78%), type B (2.4%), and type F (19.5%). However, all the *C. perfringens* isolates (n = 11) recovered from milk and milk products were grouped in toxin type A. *C. perfringens* isolates (n = 232) recovered from fish and fish products were grouped in toxin type A (96.1%) and type F (3.9%). None of the isolates of foods of animal origin were grouped under toxin types C, D, and E. However, toxinotyping for the *netB* gene (type G) was not done for any of the samples (Table 4).

The occurrence of *C. perfringens* isolates in animal origin food products was found to be concurrent with earlier study reports [72,73]. Along with the *cpa*, *cpb* and *etx* toxin genes, *C. perfringens* isolates recovered from foods of animal origin also carried *cpb2* and *cpe* toxin genes. These *cpb2* and *cpe* harboring *C. perfringens* isolates have the potential to cause food poisoning, AAD, and SD in humans after consuming contaminated food products [12].The food poisoning incidences in consumers may occur due to the lack of public awareness to adopt the hygienic practices while slaughtering, cleaning, dressing, and handling the food. In addition, personal hygiene of the food handlers matters to a great extent in food poisoning incidences.

3.1.4. *C. perfringens isolates recovered from environmental sources*

The overall prevalence of *C. perfringens* among environmental

Table 3
Prevalence and toxinotyping of *C. perfringens* isolates recovered from captive wild animals and birds.

Health status	Species	Prevalence of *C. perfringens* (%)	Toxinotypes, numbers (%)														References
			Type A		Type B	Type C				Type D			Type E	Type F		Type G	
			cpa	*cpa* + *cpb2*	(*cpa* + *cpb* + *etx*)	(*cpa* + *cpb*)	(*cpa* + *cpb* + *cpb2*)	(*cpa* + *cpb* + *cpe*)	(*cpa* + *cpb* + *cpb2*+*cpe*)	(*cpa* + *etx*)	(*cpa* + *etx* + *cpb2*)	(*cpa* + *etx* + *cpe*)	(*cpa* + *itx*)	(*cpa* + *cpe*)	(*cpa* + *cpe* + *cpb2*	(*cpa* + *netB*)	
Diseased	Wild animals	6/8 (75)	6/6 (100)	ND*	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	ND	[66]
Healthy	Wild animals	86/243 (35.4)	68/86 (79)	16/86 (18.6)	2/86 (2.3)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	[66,67]
	Wild birds	26/106 (24.5)	14/26 (53.8)	12/26 (46.1)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	[66,67]

ND*- Not done.

Table 4
Prevalence and toxinotyping of *C. perfringens* isolates recovered from foods of animal origin.

Source	Prevalence of *C. perfringens* (%)	Toxinotypes, numbers (%)														References
		Type A		Type B (*cpa* + *cpb* + *etx*)	Type C				Type D			Type E (*cpa* + *itx*)	Type F		Type G (*cpa* + *netB*)	
		cpa	*cpa*+ *cpb2*		(*cpa* + *cpb*)	(*cpa* + *cpb* + *cpb2*)	(*cpa* + *cpb* + *cpe*)	(*cpa* + *cpb* + *cpb2*+*cpe*)	(*cpa* + *etx*)	(*cpa* + *etx* + *cpb2*)	(*cpa* + *etx* + *cpe*)		(*cpa* + *cpe*)	(*cpa* + *cpe* + *cpb2*		
Meat and meat products	82/427 (19.2)	63/82 (76.8)	1/82 (1.2)	2/82 (2.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	15/82 (18.3)	1/82 (1.2)	ND*	[18,93,94]
Milk and milk products	11/55 (20)	11/11 (100)	ND	0 (0)	0 (0)	ND	0 (0)	ND	0 (0)	ND	0 (0)	0 (0)	0 (0)	ND	ND	[93]
Fish and fish products	232/1080 (21.5)	107/232 (46.1)	116/232 (50)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	9/232 (3.9)	0 (0)	ND	[95–99]

ND*- Not done.

Table 5
Prevalence and toxinotyping of *C. perfringens* isolates of environmental sources.

Prevalence of *C. perfringens* (%)	Toxinotypes, numbers (%)														References
	Type A		Type B	Type C				Type D			Type E	Type F		Type G	
	cpa	*cpa+ cpb2*	(*cpa + cpb + etx*)	(*cpa + cpb*)	(*cpa + cpb + cpb2*)	(*cpa + cpb + cpe*)	(*cpa + cpb + cpb2+cpe*)	(*cpa + etx*)	(*cpa + etx + cpb2*)	(*cpa + etx + cpe*)	(*cpa + itx*)	(*cpa + cpe*)	(*cpa + cpe + cpb2*	(*cpa + netB*)	
23/76 (30.2)	14/23 (60.8)	7/23 (30.4)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2/23 (8.7)	0 (0)	0 (0)	0 (0)	ND*	[18,100]
NA	0 (0)	0 (0)	NA**	3/13 (23.1)	10/13 (76.9)	0 (0)	0 (0)	NA	NA	NA	NA	0 (0)	0 (0)	NA	[91]

ND*- Not done; NA**- Not available.

Table 6
Prevalence and toxinotyping of *C. perfringens* isolates of human origin.

Health status	Prevalence of *C. perfringens* (%)	Toxinotypes, numbers (%)														References
		Type A		Type B	Type C				Type D			Type E	Type F		Type G	
		cpa	*cpa+ cpb2*	(*cpa + cpb + etx*)	(*cpa + cpb*)	(*cpa + cpb+ cpb2*)	(*cpa + cpb+ cpe*)	(*cpa + cpb + cpb2+cpe*)	(*cpa + etx*)	(*cpa + etx + cpb2*)	(*cpa + etx + cpe*)	(*cpa + itx*)	(*cpa + cpe*)	(*cpa + cpe + cpb2*	(*cpa + netB*)	
Diseased	70/280 (25)	39/70 (55.7)	20/70 (28.6)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	5/70 (7.1)	6/70 (8.6)	ND*	[10,18]
Healthy	36/155 (23.2)	31/36 (86.1)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	5/36 (13.9)	ND	[10,67,100]

ND*- Not done.

Table 7
Phenotypic antimicrobial resistance pattern of the *C. perfringens* isolates recovered from different sources from India.

Source	No. of isolates	Class of antibiotics[a] (% resistance)											References
		TE	MA	PEN	Q	PP	GLY	LIN	CEP	CBP	SF	CRP	
Birds, pet and domestic animals	75	20 (26.67)	30 (40)	12 (16)	9 (12)	30 (40)	5 (6.7)	4 (5.3)	ND[b]	ND	ND	ND	[11]
Humans, animals, fish and environment samples	104	47 (45.2)	ND	70 (67.3)	10 (9.6)	ND	ND	ND	57 (54.8)	ND	98 (94.2)	ND	[18]
Milk and milk products	24	15 (62.5)	12 (50)	17 (71)	ND	ND	ND	24 (100)	10 (41.7)	ND	22 (91.7)	10 (41.7)	[30]
Meat of goat, buffalo and poultry	34	10 (29.4)	ND	33 (97)	14 (41.2)	ND	ND	ND	31 (91.2)	ND	13 (38.2)	ND	[35]
Feces of chicken	30	9 (30)	ND	20 (66.6)	4 (13.3)	ND	ND	ND	18 (60)	ND	11 (36.6)	7 (23.3)	[36]
Intestinal contents of poultry	20	6 (30)	ND	2 (10)	ND	0 (0)	ND	0 (0)	ND	ND	14 (70)	6 (30)	[37]
poultry feed ingredients	30	1 (3.3)	ND	30 (100)	3 (10)	28 (93.3)	ND	4 (13.3)	4 (13.3)	ND	28 (93.3)	ND	[38]
Intestinal contents of chicken	13	6 (46.1)	ND	ND	5 (38.5)	ND	ND	0 (0)	0 (0)	ND	ND	ND	[39]
Feces of ducks	40	13 (32.5)	ND[b]	18 (45)	ND	ND	ND	ND	0 (0)	0 (0)	ND	ND	[40]
Total 370		127/370 (34.3)	42/99 (42.4)	202/357 (56.6)	45/286 (15.7)	58/125 (46.4)	5/75 (6.7)	32/162 (19.7)	120/275 (43.6)	0/40 (0)	186/242 (76.8)	23/74 (31.1)	

[a] TE- Tetracycline; MA- Macrolide; PEN- Penicillin; Q- Quinolone; PP- Polypeptides; GLY- Glycopeptides; LIN- Lincosamide; CEP- Cephalosporin; CBP- Carbapenem; SF- Sulfonamide; CRP- Chloramphenicol.
[b] ND: Not done.

samples was found to be 30.2% (23/76) (Table 5; Supplementary Table 3). Further characterization of isolates (n = 23) into different toxin types revealed that the isolates were grouped in type A (91.3%) and type D (8.7%). None of the isolates were found positive for toxin types B, C, E, and F (Table 5). However, environmental samples were not screened for the detection of the *netB* gene (type G). Additionally, *C. perfringens* isolates (n = 13) recovered from different environmental sources were grouped under toxin type C (Table 5). The presence of *C. perfringens* beta and epsilon toxin carrying type C and D strains in the environment may be associated with the causation of enteritis and enterotoxemia in sheep, cattle, goats, and humans [2,19].

3.1.5. *C. perfringens* isolates recovered from human sources

The prevalence of *C. perfringens* among humans with history of diarrhea was found to be 25% (70/280) (Table 6; Supplementary Table 4). Further categorization of isolates (n = 70) into different toxin types revealed that isolates were grouped in type A (84.3%) and type F (15.7%). The prevalence of *C. perfringens* among stool samples of healthy humans was found to be 23.2% (36/155). The isolates were grouped in toxin type A (86.1%) and type F (13.9%). None of the human isolates were found positive for toxin types B, C, D, and E (Table 6). However, human stool samples were not screened for the detection of the *netB* gene (type G).

Foodborne isolates of *C. perfringens* were found to be harboring most frequently a chromosomal enterotoxin gene (*cpe*), whereas isolates from patients having history of AAD and SD, usually have plasmid borne *cpe* gene [74]. Moreover, enteric infections caused by *C. perfringens* type F strains were observed to be more common in humans than in animals [59]. Enterotoxin-producing *C. perfringens* type A comprised around 5% of *C. perfringens* strains and was associated with foodborne illnesses [15]. However, 5–15% enterotoxin producing *C. perfringens* type A strains were involved in non-foodborne gastrointestinal diseases (AAD/SD) in humans [12,59,75]. The detection of *cpb2*-associated type F *C. perfringens* strains among healthy humans indicate that this toxin-genotype could play an important role in causing the gastrointestinal diseases in humans in adverse climatic conditions, whenever the immune system of individuals become weak [18].

3.2. Antimicrobial resistance pattern

3.2.1. Phenotypic antimicrobial resistance profile

The phenotypic antimicrobial resistance profile of *C. perfringens* isolates against the eleven classes of antibiotics revealed the highest resistance against sulfonamides (76.8%), followed by penicillin (56.6%), polypeptides (46.4%), cephalosporins (43.6%), macrolides (42.4%), tetracycline (34.3%), chloramphenicol (31.1%), lincosamides (19.7%), quinolones (15.7%), and glycopeptides (6.7%). The antibiotics of carbapenem class showed 100% sensitivity against *C. perfringens* isolates (Table 7).

Globally, there are reports of overuse or misuse of antibiotics in human and veterinary medicine without proper antibiotic susceptibility testing, which could be a major factor for increased antimicrobial resistance [33,76,77]. Moreover, the lack of awareness regarding proper antibiotic usage and lack of stewardship in many regions may further lead to emergence of antimicrobial resistance [101]. Holgel et al. [78] indicated that resistance problem in tetracycline and/or macrolides is mainly due to the transfer of the R-plasmid through conjugation among bacteria. The resistance pattern in anaerobic bacteria may be due to plasmid-mediated transferable multidrug resistance (MDR), changes in the porin molecules in the outer membrane, decreased uptake of drugs, production of beta-lactamase enzyme, inactivation of enzyme chloramphenicol acetyltransferase, changes in the antimicrobial binding receptors etc. [30,79]. Most of the poultry isolates in the present analysis were found to be resistant to multiple drugs which may be due to indiscriminate use of these antimicrobials as feed additive, in prophylaxis as well as in therapeutics in poultry industry [37,80,81].

3.2.2. Genotypic antimicrobial resistance profile

Overall, there is paucity of data on genotypic antimicrobial resistance profile of *C. perfringens* isolates from India. The only publication available reported that the maximum genotypic resistance of *C. perfringens* isolates was detected against *tet A*(P) and *tet B*(P) genes of tetracycline [41.3% (31/75)], followed by *erm*(B) and *erm*(Q) genes of erythromycin [34.6% (26/75)], *bcrA*, *bcrB*, *bcrD*, and *bcrR* genes of bacitracin [17.3% (13/75)] and *ant(3")-I* gene of gentamicin. None of the isolates were found positive for *lnu*(A) and *lnu*(B) genes of lincomycin [11].

The resistance due to tetracycline may be attributed to the presence of more than one resistance gene [i.e., *tetA*(P) and *tetB*(P)] in the isolates [48]. Higher resistance for the tetracycline has also been reported earlier in canines [82]. Similarly, Berryman et al. [83] and Soge et al. [84] reported *erm*(B) and *erm*(Q) as common macrolide resistance genes in *C. perfringens* isolates. Bacitracin resistance observed in *C. perfringens* isolates might be due to increase in the usage of this antimicrobial to improve the production in birds [85]. The resistance of *C. perfringens* isolates against aminoglycoside antibiotics might be due to its reduced intracellular transport which might have evolved as a mechanism to resist aminoglycoside antimicrobials. Hence, these antimicrobials may not be effective against *C. perfringens* [86], and also not recommended by Clinical Laboratory Standards Institute (CLSI) for antimicrobial susceptibility testing [102]. Genotypic studies on antimicrobial resistance genes of lincomycin are poorly reported and resistance might be due to as yet unknown genes [103,104].

4. Limitations and future perspectives

The present study might have some limitations. Because of the fragmented studies on *C. perfringens* from India, the true prevalence of the pathogen cannot be deduced. Hence, without knowledge of the true incidence of pathogen in the communities/regions, it is difficult to implement the mitigation strategies to control *C. perfringens*-associated animal health and public health burden. Data on occurrence and disease burden are also crucial in assessing the socio-economic encumbrance and implementation of regional and national surveillance programs. A majority of studies on *C. perfringens* in India were carried out in animal health sector, with lack of studies from human sector and human-animal-environmental interface to embrace the One Health perspective. Additionally, most of the studies were not found to follow the systematic epidemiological sampling plan and also not found to include all the genotypes (A-G) of the pathogen. Thereby, the true scenario of genotypic predominance cannot be deduced. There is also scarcity of data on antimicrobial resistance of *C. perfringens* isolates. Most of the studies on antimicrobial resistance were carried out phenotypically, and only a single study from India detected the genotypic resistance profile of *C. perfringens* isolates.

Detailed scientific studies covering different geographical areas and considering the human-animal-environment interface are need of the hour to generalize the magnitude of *C. perfringens* associated animal health and public health problems in the country.

After assessing the true prevalence of the pathogen, the development of specific virulence markers is needed to differentiate between normal and disease causing *C. perfringens* strains. User-friendly diagnostic assays need to be developed for quick detection of toxigenic strains of *C. perfringens* in leftover and/or improperly preserved and subsequently heated food products. Enhanced antimicrobial stewardship in both animal and human sectors would be a good initiative in reducing antimicrobial resistance problem in the country. Systematic studies are needed for the early detection of antibiotic resistance by targeting marker genes of *C. perfringens*. Generation of mass public health awareness will help in the prevention of food poisoning incidences due to *C. perfringens* in human population.

Further, the designing of epidemiological studies within One Health framework is strongly recommended. One Health initiative recognizes that the health of humans is interconnected to the health of animals and the environment. A principal feature of the One Health paradigm is to increase the level of coordination, collaboration, and communication between various health science professionals (veterinarians, physicians, agriculturalists etc.) and policymakers, with the aim of developing integrative policy frameworks.

5. Conclusion

The present scoping review compiled the data on *C. perfringens* recovered from humans, animals, animal-based foods, and associated environment samples from India in the last two decades. The pathogen was detected from all the targeted sources with toxin type A as the most prevalent genotype. Along with the alpha toxin gene (*cpa*), beta (*cpb*), epsilon (*etx*), iota (*itx*), enterotoxin (*cpe*), beta2 toxin (*cpb2*) and NetB (*netB*) toxins were also detected in different combinations. The enterotoxin and beta2 toxins are involved in gastrointestinal diseases in humans and animals. However, the presence of the *cpe* and *cpb2* genes in healthy animals does not necessarily indicate the disease condition of individuals, but they may act as carrier of this pathogen to other humans and animals. Antimicrobial sensitivity profile of *C. perfringens* isolates recovered from different sources reveal that carbapenem and lincomycin class of antibiotics was found to be most effective against *C. perfringens.* We recommend detailed systematic epidemiological studies on *C. perfringens* covering large geographical areas and considering the human-animal-environment interface to generalize the real magnitude of problem in India for establishing a reliable database.

Declaration of competing interest

The authors declare that there is no conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.anaerobe.2022.102639.

References

[1] B.A. McClane, S.L. Robertson, J. Li, Clostridium perfringens, in: M.P. Doyle, R.L. Buchanan (Eds.), In: Food Microbiology: Fundamentals and Frontiers, fourth ed., ASM press, Washington D.C., 2013, pp. 465–489.
[2] I. Mehdizadeh Gohari, A. Navarro, M. Li, J. Shrestha, A. Uzal, F, A. McClane, B, Pathogenicity and virulence of Clostridium perfringens, Virulence 12 (1) (2021) 723–753.
[3] N.A. Ezzeldeen, A.M. Ammar, B. Shalaby, M.E. Haririr, W.S. Omar, Rapid detection of Clostridium perfringens in seafood, Adv. Environ. Biol. 10 (4) (2016) 174–182.
[4] J.P. Yadav, S.C. Das, P. Dhaka, A.K. Mukhopadhyay, G. Chowdhury, S. Naskar, S.S. Malik, Pulsed-field gel electrophoresis of enterotoxic Clostridium perfringens type A isolates recovered from humans and animals in Kolkata, India, Int. J. Veterinary Sci. Med. 6 (1) (2018) 123–126.
[5] M. Lindström, A. Heikinheimo, P. Lahti, H. Korkeala, Novel insights into the epidemiology of Clostridium perfringens type A food poisoning, Food Microbiol. 28 (2) (2011) 192–198.
[6] A.A.P. Milton, K.M. Momin, G.B. Priya, S. Ghatak, P.N. Gandhale, M. Angappan, S. Das, A. Sen, A novel in situ methodology for visual detection of Clostridium perfringens in pork harnessing saltatory rolling circle amplification, Anaerobe 69 (2021), 102324.
[7] S.B. Melville, L. Craig, Type IV pili in Gram-positive bacteria, Microbiol. Mol. Biol. Rev. 77 (3) (2013) 323–341.
[8] R. Kiu, L.J. Hall, An update on the human and animal enteric pathogen Clostridium perfringens, Emerg. Microb. Infect. 7 (1) (2018) 1–15.
[9] J.I. Rood, V. Adams, J. Lacey, D. Lyras, B.A. McClane, S.B. Melville, R.J. Moore, M.R. Popoff, M.R. Sarker, J.G. Songer, F.A. Uzal, Expansion of the *Clostridium perfringens* toxin-based typing scheme, Anaerobe 53 (2018) 5–10.
[10] K. Anju, K. Karthik, V. Divya, M.L.M. Priyadharshini, R.K. Sharma, S. Manoharan, Toxinotyping and molecular characterization of antimicrobial resistance in Clostridium perfringens isolated from different sources of livestock and poultry, Anaerobe 67 (2021), 102298.
[11] L. Joshy, R. Chaudhry, B. Dhawan, L. Kumar, B.K. Das, Incidence and characterization of Clostridium perfringens isolated from antibiotic-associated diarrhoeal patients: a prospective study in an Indian hospital, J. Hosp. Infect. 63 (3) (2006) 323–329.
[12] D.J. Fisher, K. Miyamoto, B. Harrison, S. Akimoto, M.R. Sarker, B.A. McClane, Association of beta2 toxin production with *Clostridium perfringens* type A human gastrointestinal disease isolates carrying a plasmid enterotoxin gene, Mol. Microbiol. 56 (3) (2005) 747–762.
[13] J. Sasaki, M. Goryo, M. Asahina, M. Makara, S. Shishido, K. Okada, Hemorrhagic enteritis associated with Clostridium perfringens type A in a dog, J. Vet. Med. Sci. 61 (2) (1999) 175–177.
[14] J.G. Smedley, B.A. McClane, Fine mapping of the N-terminal cytotoxicity region of *Clostridium perfringens* enterotoxin by site-directed mutagenesis, Infect. Immun. 72 (2004) 6914–6923.
[15] F.A. Uzal, J.C. Freedman, A. Shrestha, J.R. Theoret, J. Garcia, M.M. Awad, V. Adams, R.J. Moore, J.I. Rood, B.A. McClane, Towards an understanding of the role of *Clostridium perfringens* toxins in human and animal disease, Future Microbiol. 9 (3) (2014) 361–377.
[16] R.L. Scharff, Economic burden from health losses due to foodborne illness in the United States, J. Food Protect. 75 (1) (2012) 123–131.
[17] J.E. Grass, L.H. Gould, B.E. Mahon, Epidemiology of food borne disease outbreaks caused by *Clostridium perfringens*, United States, 1998-2010, Food borne Pathog. Dis. 10 (2) (2013) 131–136.
[18] J.P. Yadav, S.C. Das, P. Dhaka, D. Vijay, M. Kumar, A.K. Mukhopadhyay, G. Chowdhury, P. Chauhan, R. Singh, K. Dhama, S.V.S. Malik, Molecular characterization and antimicrobial resistance profile of Clostridium perfringens type A isolates from humans, animals, fish and their environment, Anaerobe 47 (2017) 120–124.
[19] F.A. Uzal, S.S. Diab, P. Blanchard, J. Moore, L. Anthenill, F. Shahriar, *Clostridium perfringens* type C and *Clostridium difficile* co-infection in foals, Vet. Microbiol. 156 (2012) 395–402.
[20] A. Banaszkiewicz, J. Kądzielska, A. Gawrońska, H. Pituch, P. Obuch-Woszczatyński, P. Albrecht, G. Młynarczyk, A. Radzikowski, Enterotoxigenic Clostridium *perfringens* infection and pediatric patients with inflammatory bowel disease, J. Crohn's Colitis. 8 (2014) 276–281.
[21] J.C. Freedman, A. Shrestha, B.A. McClane, *Clostridium perfringens* enterotoxin: action, genetics, and translational applications, Toxins 8 (3) (2016) 73.
[22] C.K. Athira, A.A.P. Milton, A. Reddy, A.M. Rajendrakumar, M.R. Verma, A. Kumar, V.K. Nagaleekar, R.K. Agarwal, Diversity of toxin-genotypes among Clostridium perfringens isolated from healthy and diarrheic neonatal cattle and buffalo calves, Anaerobe 49 (2018) 99–102.
[23] J.G. Songer, Clostridial enteric diseases of domestic animals, Clin. Microbiol. Rev. 9 (2) (1996) 216.
[24] F.F. Siqueira, M.O. Almeida, T.M. Barroca, C.C. Horta, A.O. Carmo, R.O. Silva, P.S. Pires, F.C. Lobato, E. Kalapothakis, Characterization of polymorphisms and isoforms of the Clostridium perfringens phospholipase C gene (plc) reveals high genetic diversity, Vet. Microbiol. 159 (3–4) (2012) 397–405.
[25] F.A. Uzal, J.E. Vidal, B.A. McClane, A.A. Guraj, *Clostridium perfringens* toxins involved in mammalian veterinary diseases, Open Toxicol. J. 2 (2014) 24–42.
[26] G.B. Priya, R.K. Agrawal, A.A.P. Milton, M. Mishra, S.K. Mendiratta, R.K. Agarwal, A. Luke, B.R. Singh, D. Kumar, Development and evaluation of isothermal amplification assay for the rapid and sensitive detection of Clostridium perfringens from chevon, Anaerobe 54 (2018) 178–187.
[27] X. Zhao, C.W. Lin, J. Wang, D.H. Oh, Advances in rapid detection methods for foodborne pathogens, J. Microbiol. Biotechnol. 24 (3) (2014) 297–312.
[28] K.M. Osman, Y.A. Soliman, Z.M.S. Amin, M.A.K. Aly, Prevalence of *Clostridium perfringens* type A isolates in commercial broiler chickens and parent broiler breeder hens in Egypt, Rev. Sci. Tech. OIE 31 (2012), 931e941.
[29] M. Teuber, Spread of antibiotic resistance with food-borne pathogens, Cell. 412 Mol. Life Sci. 56 (1999), 755e763.
[30] R.V. Singh, B. Bist, Antimicrobial profile of Clostridium perfringens isolates from dairy products, J. Anim. Res. 3 (2) (2013) 147.
[31] I. Phillips, M. Casewell, T. Cox, B. De Groot, C. Friis, R. Jones, C. Nightingale, R. Preston, J. Waddell, Does the use of antibiotics in food animals pose a risk to human health? A critical review of published data, J. Antimicrob. Chemother. 53 (1) (2004) 28–52.
[32] G.D. Wright, Q&A: antibiotic resistance: where does it come from and what can we do about it? BMC Biol. 8 (1) (2010) 1–6.
[33] E.M. Wellington, A.B. Boxall, P. Cross, E.J. Feil, W.H. Gaze, P.M. Hawkey, A.S. Johnson-Rollings, D.L. Jones, N.M. Lee, W. Otten, C.M. Thomas, The role of the natural environment in the emergence of antibiotic resistance in Gram-negative bacteria, Lancet Infect. Dis. 13 (2) (2013) 155–165.
[34] L. Garcia-Migura, R.S. Hendriksen, L. Fraile, F.M. Aarestrup, Antimicrobial resistance of zoonotic and commensal bacteria in Europe: the missing link between consumption and resistance in veterinary medicine, Vet. Microbiol. 170 (1–2) (2014) 1–9.
[35] R.V. Singh, K.N. Bhilegaonkar, R.K. Agarwal, Studies on occurrence and characterization of Clostridium perfringens from select meats, J. Food Saf. 25 (2) (2005) 146–156.
[36] A. Agarwal, G. Narang, N.K. Rakha, N.K. Mahajan, Anshu Sharma, In vitro lecithinase activity and antibiogram of clostridium perfringens isolated from broiler chickens, Haryana Vet. 48 (2009) 81–84.
[37] S. Malmarugan, A. Boobalan, N. Dorairajan, Necrotic Enteritis in broiler and layer farms in Tamil Nadu, India, Int. J. Agro Vet. Med. Sci. (IJAVMS) 6 (4) (2012) 241–249.
[38] S. Udhayavel, G.T. Ramasamy, V. Gowthaman, S. Malmarugan, K. Senthilvel, Occurrence of *Clostridium perfringens* contamination in poultry feed ingredients: isolation, identification and its antibiotic sensitivity pattern, Animal Nutr. 3 (3) (2017) 309–312.
[39] H. Sarmah, R. Hazarika, P. Deka, R. Sharma, Isolation and characterization of *Clostridium perfringens* from suspected cases of necrotic enteritis (NE) in broiler chicken, Int J Livestock Res 9 (8) (2019) 221–232.
[40] A.A.P. Milton, R. Sanjukta, A.P. Gogoi, K.M. Momin, G.B. Priya, S. Das, S. Ghatak, A. Sen, B.K. Kandpal, Prevalence, molecular typing and antibiotic resistance of Clostridium perfringens in free range ducks in Northeast India, Anaerobe 64 (2020), 102242.
[41] H. Arksey, L. O'Malley, Scoping studies: towards a methodological framework, Int. J. Soc. Res. Methodol. 8 (1) (2005) 19–32.
[42] D. Levac, H. Colquhoun, K.K. O'Brien, Scoping studies: advancing the methodology, Implement. Sci. 5 (1) (2010) 1–9.
[43] A.C. Tricco, E. Lillie, W. Zarin, K.K. O'Brien, H. Colquhoun, D. Levac, D. Moher, M.D. Peters, T. Horsley, L. Weeks, S. Hempel, PRISMA extension for scoping

reviews (PRISMA-ScR): checklist and explanation, Ann. Intern. Med. 169 (7) (2018) 467–473.
[44] P.S. Dar, S.A. Wani, A.H. Wani, I. Hussain, R. Maqbool, M.Y. Ganaie, Z.A. Kashoo, S. Qureshi, Isolation, identification and molecular characterization of Clostridium perfringens from poultry in Kashmir valley, India, J. Entomol. Zool. Stud 5 (5) (2017) 409–414.
[45] S. Nazki, S.A. Wani, R. Parveen, S.A. Ahangar, Z.A. Kashoo, S. Hamid, Z.A. Dar, T.A. Dar, P.A. Dar, Isolation, molecular characterization and prevalence of Clostridium perfringens in sheep and goats of Kashmir Himalayas, India, Vet. World 10 (12) (2017) 1501.
[46] S. Shoukat, Z.H. Munshi, S.A. Wani, M. Ali, Z.A. Kashoo, S.A. Shah, H. Wani, U. Ali, S. Bukhari, Isolation, prevalence and molecular toxinotyping of Clostridium perfringens from healthy and diseased sheep of Kashmir, India, J. Entomol. Zool. Stud 6 (6) (2018) 805–810, 2018.
[47] B.E. Engstrom, C. Fermer, A. Saarinen, V. Baverud, A. Gunnarsson, Molecular typing of isolates of Clostridium perfringens from healthy and diseased poultry, Vet. Microbiol. 94 (2003) 225–235.
[48] A. Gholamiandehkordi, V. Eeckhaut, A. Lanckriet, L. Timbermont, L. Bjerrum, R. Ducatelle, F. Haesebrouck, F. Van Immerseel, Antimicrobial resistance in Clostridium perfringens isolates from broilers in Belgium, Vet. Res. Commun. 33 (2009) 1031–1037.
[49] M. Ezatkhah, M. Alimolaei, N. Shahdadnejad, The prevalence of netB gene in isolated Clostridium perfringens from organic broiler farms suspected to necrotic enteritis, Int.J. Enteric Pathog. 4 (3) (2016), e35667.
[50] A.M. Selim, M.M. Elhaig, I. Zakaria, A. Ali, Bacteriological and molecular studies of *Clostridium perfringens* infections in newly born calves, Trop. Anim. Health Prod. 49 (2017) 201–205.
[51] J.W. Chon, K.H. Seo, D. Bae, J.H. Park, S. Khan, K. Sung, Prevalence, toxin gene profile, antibiotic resistance, and molecular characterization of *Clostridium perfringens* from diarrheic and non-diarrheic dogs in Korea, J. Vet. Sci. 19 (2018) 368–374.
[52] S. Sayeed, F.A. Uzal, D.J. Fischer, Beta toxin is essential for the intestinal virulence of Clostridium perfringes type C disease isolate CN3685 in a rabbit ileal loop model, Mol. Microbiol. 67 (2006) 15–30.
[53] A.C. Mignaqui, R.B. Marcellino, T. Ronco, J.S. Pappalardo, B. Nonnemann, K. Pedersen, C.A. Robles, Isolation and molecular characterization of Clostridium perfringens from healthy Merino lambs in Patagonia region, Argentina, Anaerobe 43 (2017) 35–38.
[54] C.G. Baums, U. Schotte, G. Amtsberg, R. Goethe, Diagnostic multiplex PCR for toxin genotyping of Clostridium perfringens isolates, Vet. Microbiol. 100 (1–2) (2004) 11–16.
[55] K. Karthik, K. Manimaran, R. Bharathi, K. Shoba, Report of enterotoxaemia in goat kids, Adv. Anim. Vet. Sci. 5 (2017) 289–292.
[56] H. Kalender, H.B. Ertas, B. Cetinkaya, A. Muz, N. Arslan, A. Kilic, Typing of isolates of Clostridium perfringens from healthy and diseased sheep by multiplex PCR, Veterinarni Medicina-Praha- 50 (10) (2005) 439.
[57] J.Y. Park, S. Kim, J.Y. Oh, H.R. Kim, I. Jang, H.S. Lee, et al., Characterization of Clostridium perfringens isolates obtained from 2010 to 2012 from chickens with necrotic enteritis in Korea, Poultry Sci. 94 (2015) 1158–1164.
[58] A.R. Jabbari, F. Tekyei, M. Esmaelizad, L.R. Pilehchian, Occurrence of Beta2 toxigenic Clostridium perfringens isolates with different toxin types in Iran, Arch. Razi Inst. 67 (2) (2012) 133–137.
[59] F.A. Uzal, M.A. Navarro, J. Li, J.C. Freedman, A. Shrestha, B.A. McClane, Comparative pathogenesis of enteric clostridial infections in humans and animals, Anaerobe 53 (2018) 11–20.
[60] F. Alsheikhly, R.B. Truscott, Pathology of necroticenteritis of chickens following infusion of broth cultures of *Clostridium perfringens* into duodenum, Avian Dis. 21 (1977) 241–255.
[61] R.V. Titball, C.E. Naylor, A.K. Basak, The *Clostridium perfringens* alpha-toxin, Anaerobe 5 (1999) 51–64.
[62] A.L. Keyburn, T.L. Bannam, R.J. Moore, J.I. Rood, NetB, pore-forming toxin from necrotic enteritis strains of *Clostridium perfringens*, Toxins 2 (7) (2010) 1913–1927.
[63] M.A. Bailey, K.S. Macklin, J.T. Krehling, Use of a Multiplex PCR for the Detection of Toxin-Encoding Genes netB and tpeL in Strains of *Clostridium perfringens*, vol. 2013, ISRN Veterinary Science, 2013, pp. 1–4, 2013.
[64] h. To, T. Suzuki, F. Kawahara, K. Uetsuka, S. Nagai, T. Nunoya, Experimental induction of necrotic enteritis in chickens by a netB-positive Japanese isolate of *Clostridium perfringens*, J. Vet. Med. Sci. 79 (2) (2017) 350–358.
[65] P. Shelke, M. Pawade, P. Mhase, C. Mote, V. Lonkar, D. Suryawanshi, J. Sangle, Toxinotyping of *Clostridium perfringens* in poultry from necrotic enteritis cases, J. Anim. Res. 10 (1) (2020) 149–155.
[66] M.S. Rahman, R.K. Sharma, P. Borah, A. Chakraborty, M.R. Devi, N. Longjam, Characterization of Clostridium perfringens isolated from mammals and birds from Guwahati city, India, J. Venom. Anim. Toxins Incl. Trop. Dis. 18 (1) (2012) 83–87.
[67] A.A.P. Milton, R.K. Agarwal, G.B. Priya, M. Saminathan, M. Aravind, A. Reddy, C.K. Athira, T. Ramees, A.K. Sharma, A. Kumar, Prevalence and molecular typing of Clostridium perfringens in captive wildlife in India, Anaerobe 44 (2017) 55–57.
[68] S. Fujita, T. Kageyama, Polymerase chain reaction detection of Clostridium perfringens in feces from captive and wild chimpanzees, Pan troglodytes, J. Med. Primatol. 36 (1) (2007) 25–32.
[69] J. Razmyar, G.A. Kalidari, A. Tolooe, M. Rad, A.R. Movassaghi, Genotyping of Clostridium perfringens isolated from healthy and diseased ostriches (Struthio camelus), Iran. J. Microbiol. 6 (1) (2014) 31.
[70] R.O.S. Silva, M.L. D'Elia, É.P.T. Teixeira, P.L.L. Pereira, D.F. de Magalhães Soares, Á.R. Cavalcanti, A. Kocuvan, M. Rupnik, A.L.Q. Santos, C.A.O. Junior, F.C.F. Lobato, Clostridium difficile and Clostridium perfringens from wild carnivore species in Brazil, Anaerobe 28 (2014) 207–211.
[71] R.O. Silva, L.R. Almeida, C.A.O. Junior, P.C. Lima, D.F. Soares, P.L. Pereira, I.J. Silva, F.C. Lobato, Isolation and genotyping of clostridium perfringens from free-living south american coati (nasua nasua), J. Zoo Wildl. Med. 47 (1) (2016) 333–336.
[72] Q. Wen, B.A. McClane, Detection of enterotoxigenic *Clostridium perfringens* type A isolates in American retail foods, Appl. Environ. Microbiol. 70 (5) (2004) 2685–2691.
[73] I.A. Elham, A.A. Nahla, Incidence of *Clostridium perfringens* in meat products at some Egyptian governorates, Int. J. Microbiol. 2 (3) (2011) 196–203.
[74] J. Li, V. Adams, T.L. Bannam, K. Miyamoto, J.P. Garcia, F.A. Uzal, J.I. Rood, B.A. McClane, Toxin plasmids of *Clostridium perfringens*, Microbiol. Mol. Biol. Rev. 77 (2) (2013) 208–233.
[75] Y.J. Kim, S.H. Kim, J. Ahn, S. Cho, D. Kim, K. Kim, H. Lee, H. Son, H.J. Lee, D. Yong, J.Y. Choi, Prevalence of Clostridium perfringens toxin in patients suspected of having antibiotic-associated diarrhea, Anaerobe 48 (2017) 34–36.
[76] J.A. Ayukekbong, M. Ntemgwa, A.N. Atabe, The threat of antimicrobial resistance in developing countries: causes and control strategies, Antimicrob. Resist. Infect. Control 6 (1) (2017) 1–8.
[77] U.O. Adekanye, A.B. Ekiri, E. Galipó, A.B. Muhammad, A. Mateus, R.M. La Ragione, A. Wakawa, B. Armson, E. Mijten, R. Alafiatayo, G. Varga, Knowledge, attitudes and practices of veterinarians towards antimicrobial resistance and stewardship in Nigeria, Antibiotics 9 (8) (2020) 453.
[78] C.S. Hölzel, K.S. Harms, K. Schwaiger, J. Bauer, Resistance to linezolid in a porcine Clostridium perfringens strain carrying a mutation in the rplD gene encoding the ribosomal protein L4, Antimicrob. Agents Chemother. 54 (3) (2010) 1351–1353.
[79] S.M. Fingold, Mechanism of resistance in anaerobic bacteria and new developments in testing, Diagn. Microbiol. Infect. Dis. 12 (4) (1989) 117s–120s.
[80] M.S. Diarra, F. Molouin, Antibiotic in Canadian poultry productions and anticipated alternatives, Front. Microbiol. 5 (2014) 282, https://doi.org/10.3389/fmicb.2014.00282.
[81] S. Mwangi, J. Timmons, S. Fitz-Coy, S. Parveen, Characterization of Clostridium perfringens recovered from broiler chicken affected by necrotic enteritis, Poultry Sci. 98 (1) (2019) 128–135.
[82] E.J. Kather, S. L Marks, J.E. Foley, Determination of the prevalence of antimicrobial resistance genes in canine Clostridium perfringens isolates, Vet. Microbiol. 113 (2006) 97–101.
[83] D.I. Berryman, M. Lyristis, J.I. Rood, Cloning and sequence analysis of ermQ, the predominant macrolidelincosamide-streptogramin B resistance gene in Clostridium perfringens, Antimicrob. Agents Chemother. 38 (1994) 1041–1046.
[84] O.O. Soge, L.D. Tivoli, J.S. Meschke, M.C. Roberts, A conjugative macrolide resistance gene, mef(A), in environmental Clostridium perfringens carrying multiple macrolide and/or tetracycline resistance genes, J. Appl. Microbiol. 106 (2009) 34–40.
[85] X. Han, X.D. Du, L. Southey, D.M. Bulach, T. Seemann, X.X. Yan, T.L. Bannam, J.I. Rood, Functional analysis of a bacitracin resistance determinant located on ICECp1, a novel Tn916-like element from a conjugative plasmid in Clostridium perfringens, Antimicrob. Agents Chemother. 59 (2015) 6855–6865.
[86] L.E. Bryan, S.K. Kowand, H.M. Van Den Elzen, Mechanism of aminoglycoside antibiotic resistance in anaerobic bacteria: Clostridium perfringens and Bacteroides fragilis, Antimicrob. Agents Chemother. 15 (1979) 7–13.
[87] N.V. Kumar, D. Sreenivasulu, Y.N. Reddy, Prevalence of Clostridium perfringens toxin genotypes in enterotoxemia suspected sheep flocks of Andhra Pradesh, Vet. World 7 (12) (2014) 1132–1136.
[88] N.V. Kumar, D. Sreenivasulu, Y.N. Reddy, Identification of Clostridium Perfringens Type C and D from enterotoxemia suspected lambs in Andhra Pradesh, Indian Vet. J. 93 (2016) 29–31, 01.
[89] B. Radhika, N.V. Kumar, D. Sreenivasulu, Detection of Clostridium perfringens alpha toxin gene in lambs by loop mediated isothermal amplification, Vet. World 9 (1) (2016) 60.
[90] Z.A. Kashoo, S.A. Wani, A.H. Wani, S.M. Khan, S. Qureshi, I. Hussain, S. Farooq, J.A. Malla, Molecular characterisation of Clostridium perfringens type D isolated from sheep in Kashmir himalayas, India, Int. J. Curr. Microbiol. App. Sci 7 (8) (2018) 3682–3688.
[91] S. Das, S. Majumder, C. Mathur, J.J. Kingston, Molecular characterization and phylogenetic analysis of Clostridium perfringens from animals and their environments by cpn60 UT sequencing analysis, Infect. Genet. Evol. 58 (2018) 209–217.
[92] N.P. Kumar, N.V. Kumar, A. Karthik, Molecular detection and characterization of Clostridium perfringens toxin genes causing necrotic enteritis in broiler chickens, Trop. Anim. Health Prod. 51 (6) (2019) 1559–1569.
[93] S. Haque, R.K. Sharma, N.N. Barman, L.B. Devi, L.M. Barkalita, I. Hussain, Characterization of enterotoxin producing Clostridium perfringens isolated from foods of animal origin, Indian J. Anim. Res. 52 (1) (2018) 111–115.
[94] S.A. Hussain, M.M. Willayat, M.A. Rather, Enterotoxigenecity and typing of Clostridium perfringens isolated from raw meat, meat products and water samples in Kashmir, India, Int. J. Curr. Microbiol. App. Sci 7 (8) (2018) 480–489.

[95] D. Arunava, J. Adarsh, Clostridium perfringens type A from fresh water fishes, Int. J. Adv. Biotechnol. Res. 3 (2012) 680–687.
[96] A. Das, A. Jain, Genotyping of Clostridium perfringens from fresh water fish and fish pickles, J. Microbiol. Biotechnol. Food Sci. 2 (1) (2012) 162–174.
[97] J.P. Yadav, S.C. Das, P. Dhaka, D. Vijay, M. Kumar, P. Chauhan, R. Singh, K. Dhama, S.V.S. Malik, A. Kumar, Isolation, genotyping and antibiogram profile of Clostridium perfringens isolates recovered from freshwater fish and fish products from Kolkata region, J. Pure Appl. Microbiol. 10 (4) (2016) 2807–2814.
[98] A. Poluru, R. Sahu, K. Nelapati, Detection and characterization of *Clostridium perfringens* in seafoods, Int J Livestock Res 7 (7) (2017) 177–183.
[99] N. Wani, S.A. Wani, Z.H. Munshi, S. Ahmad, M.R. Shah, A. Hussain, Z. Kashoo, N. Nabi, Isolation and virulence gene profiling of Clostridium perfringens from freshwater fish, J. Entomol. Zool. Stud 6 (3) (2018) 176–181.
[100] A.K. Verma, M.Y. Abdel-Glil, A. Madesh, S. Gupta, A.C. Karunakaran, S. Inbaraj, V.K. Nagaleekar, P. Chaudhuri, R.K. Agarwal, P. Thomas, Multilocus sequence typing of Clostridium perfringens strains from neonatal calves, dairy workers and associated environment in India, Anaerobe 63 (2020), 102212.
[101] Z. Haider, T. Ali, A. Ullah, A. Basit, H. Tahir, H. Tariq, S.Z. Ilyas, Z. Hayat, S.U. Rehman, Isolation, toxinotyping and antimicrobial susceptibility testing of *Clostridium perfringens* isolated from Pakistan poultry, Anaerobe 73 (2022), 102499.
[102] CLSI, Performance standards for antimicrobial susceptibility testing, in: CLSISupplementM100. Wayne, PA: Clinical and Laboratory Stand-ardsInstitute, Pennsylvania: USA, 30th ed., 2020.
[103] A. Martel, L.A. Devriese, K. Cauwerts, K. De Gussem, A. Decostere, F. Haesebrouck, Susceptibility of *Clostridium perfringens* strains from broiler chicken to anbiotics and anticoccidials, Avian Pathol. 33 (2004) 3–7.
[104] R.O.S. Silva, F.M. Salvarani, R.A. Assis, N.R.S. Martins, P.S. Pires, F.C.F. Lobato, Antimicrobial susceptibility of *Clostridium perfringens* strains isolated from broiler chickens, Braz. J. Microbiol. 40 (2009) 262–264.